

CURIONEST CHEMISTRY

CHEMISTRY FOUNDATIONS

Lab Safety, Equipment & Measurement



GHS safety



Graduated cylinder



Electronic balance



Erlenmeyer flask

PRINT + EDITABLE

Teacher guide | visual reference pages | stations | practical | quiz | full key

Grades 9-11 • 3-5 class periods • US Letter

START HERE

Teacher Guide

A flexible 3-5 day launch unit for Grades 9-11

PURPOSE Students build safe lab habits, recognize essential equipment, and make defensible measurements before the first wet lab.

Day	Focus	Suggested evidence
1	Safety norms + Safety Detective	Correct unsafe actions using evidence
2	GHS labels + SDS navigation	Locate hazard, PPE, first-aid information
3	Equipment identification + tool choice	Match tool to task and justify
4	Measurement stations	Read meniscus; tare and report mass
5	Practical + quiz	Demonstrate procedures and explain choices

Prep checklist

- Print pages labeled Student Page or Assessment; retain Teacher Page and Teacher Key pages.
- Set out clean, empty equipment only. Do not use chemicals for the identification stations.
- Test the eyewash and safety shower according to school policy; never activate them solely as a student demonstration.
- Use local emergency procedures and the SDS for every chemical used in later labs.

STANDARDS ROLE This foundational skills unit supports NGSS science practices such as planning investigations, analyzing data, and using mathematics. It is not presented as a standalone HS-PS1 performance-expectation unit.

Suggested materials

Goggles, lab apron, beaker, Erlenmeyer flask, graduated cylinder, test tube and rack, dropper, funnel, electronic balance, wash bottle, ring stand, hot plate, and printed station cards.

SAFETY NOTE This resource supports instruction; it does not replace district policy, teacher training, equipment inspection, or chemical-specific SDS review.

Lab Safety Detective

Read eight classroom snapshots. Explain the safer action for each.

LAB SAFETY DETECTIVE

Eight classroom snapshots • Identify the unsafe choice before writing the safer action.

1 EYE PROTECTION

A student begins work with splash goggles resting on the forehead.

What should happen instead?

2 FOOD & DRINK

A drink cup sits beside the lab notebook and glassware.

What should happen instead?

3 LABELING

A beaker contains blue liquid but has no chemical label.

What should happen instead?

4 HEAT

Loose paper is placed beside a lit laboratory burner.

What should happen instead?

5 BROKEN GLASS

Broken glass is left on the bench after cleanup begins.

What should happen instead?

6 HOUSEKEEPING

A backpack blocks the aisle beside the lab station.

What should happen instead?

7 CHEMICALS

An open reagent bottle is left unattended on the bench.

What should happen instead?

8 PERSONAL SAFETY

Long hair hangs loose near the work area and heat source.

What should happen instead?

#	Unsafe choice	Safer action
1		
2		
3		
4		
5		
6		
7		
8		

GHS Pictogram Reference

Symbols communicate hazard classes; they do not replace the full label or SDS.

GHS Hazard Pictograms

 Exploding bomb Explosives; some self-reactives/peroxides	 Flame Flammable and related fire hazards	 Oxidizer Oxidizing gases, liquids, or solids
 Gas cylinder Gases under pressure	 Corrosion Skin burns/eye damage; corrosive to metals	 Skull Fatal or toxic acute toxicity
 Exclamation Irritant or harmful effects	 Health hazard Cancer, organ, respiratory, aspiration hazards	 Environment Aquatic toxicity; not OSHA-mandated

REMEMBER OSHA requires eight pictograms in its Hazard Communication Standard. The environmental pictogram is shown because students may see it, but OSHA does not mandate it.

STUDENT PAGE 5

Equipment Reference

Study the name and function. Circle tools available in your classroom.

Lab Equipment Reference - Set A

**Beaker**

Holds/mixes liquids; approximate volume

**Erlenmeyer flask**

Swirls mixtures with less splashing

**Graduated cylinder**

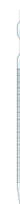
Measures liquid volume accurately

**Test tube**

Runs small-scale reactions

**Buret**

Delivers precise variable volumes

**Dropper pipet**

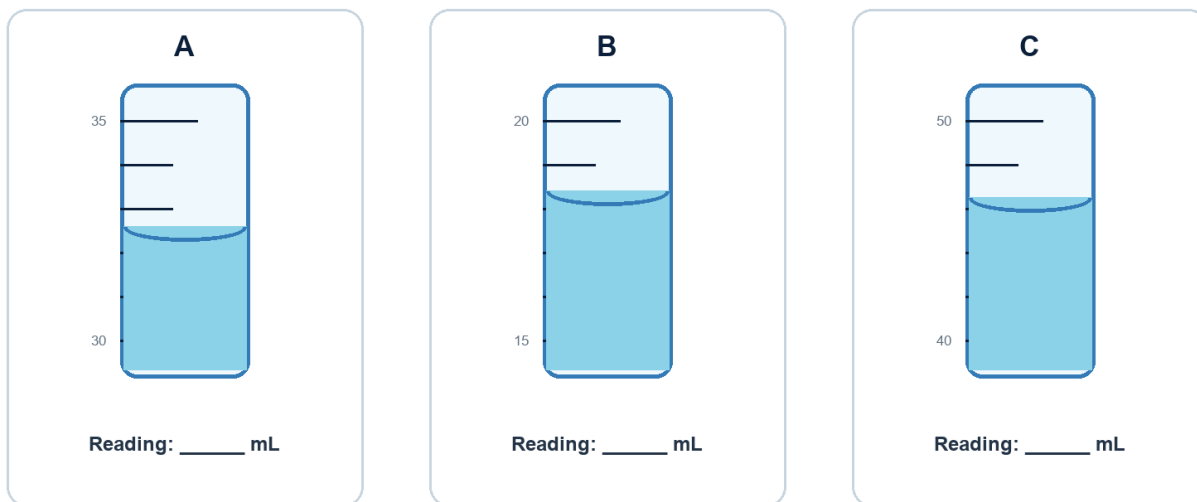
Transfers small amounts dropwise

STUDENT PAGE 10

Meniscus and Precision Practice

Read the bottom of each meniscus at eye level.

Read Each Graduated Cylinder



1. Which cylinder permits the most precise reading? Explain using the scale divisions.

2. A student records Cylinder A as 33 mL. What important measurement information is lost?

3. Describe one source of parallax error and how to prevent it.

ASSESSMENT

Lab Skills Practical

Teacher observation rubric; score each criterion 0-2.

Criterion	2 - Meets	1 - Developing	0 - Not yet
PPE and readiness	Correct PPE; area clear; reads task first	Minor prompt needed	Unsafe/not ready
Tool identification	Names tool and appropriate use	Name or use incomplete	Incorrect
Volume technique	Eye level; bottom meniscus; unit	One technique error	Multiple errors
Balance technique	Tares; waits for stable reading; unit	One technique error	Multiple errors
Explanation	Uses safety/precision evidence	Partial reason	No defensible reason

Score: _____ / 10 Teacher feedback: _____

REASSESSMENT Allow a second attempt after feedback. The goal is safe, repeatable performance—not a one-time score.

TEACHER KEY 1

Answer Key - Safety and GHS

Suggested responses; accept equivalent scientifically sound wording.

Item	Suggested answer
Safety Detective 1-4	1 goggles over eyes; 2 remove drink; 3 label container; 4 move paper/flammables away and control flame.
Safety Detective 5-8	5 notify teacher; use broken-glass bin with proper tools; 6 remove backpack from aisle; 7 do not leave a reagent container open or unlabeled; 8 secure loose hair.
Label Q1	Flame = flammable hazard; exclamation = irritant/harmful effects.
Label Q2	Any two: keep away from ignition; wear eye protection; use adequate ventilation.
Label Q3-4	First aid = SDS Section 4. Exposure controls/PPE = Section 8.
Label Q5	The label is a quick warning; the SDS provides detailed controls, first aid, storage, stability, and disposal information.

Readiness explanations

Goggles must cover the eyes to protect against splashes. Chemicals are not returned to stock because they may contaminate the original reagent.